

Build a Hybrid Cloud with Eucalyptus

Eucalyptus is open source software for building an AWS (Amazon Web Services) compatible private and hybrid cloud computing environment. It provides flexibility, cost benefits, agility and data governance.



Eucalyptus is open source software that implements an Amazon Web Services compatible cloud, which is cost-effective, flexible and secure. It can be easily deployed in existing IT infrastructures to enjoy the benefits of both public and private cloud models. Eucalyptus is an acronym for 'Elastic Utility Computing Architecture for Linking Your Programs to Useful Systems'. Basically, Eucalyptus provides an Infrastructure as a Service (IaaS) offering. The main advantage is that it provides easy and secure deployment. The private cloud is deployed in the premises of the enterprise and can be accessed by users over the intranet, so critical and important data remains secure from outside intrusion. Also, it provides AWS APIs. So, at any time, consumers can easily migrate or load balance their less sensitive data into the Amazon public cloud; thus, they don't have to worry about the elasticity of their network.

History

Development on Eucalyptus began as a research project in US based Rice University in 2003. In 2009, a company named Eucalyptus Systems was formed to commercialise Eucalyptus software. Later, in 2012, the firm entered into an agreement with AWS for maintaining compatibility and API support. In 2014, it was acquired by HP (Hewlett-Packard), which

incidentally has its own cloud offerings under the HPE Helion banner. The Helion portfolio has a variety of cloud related products, which includes HP's own flavour of OpenStack called HP Helion OpenStack. Now, Eucalyptus is a part of the HPE portfolio and is called HPE Helion Eucalyptus. It provides an open solution for building a hybrid cloud, leveraging the benefits of other HP Helion products.

Eucalyptus architecture

Figure 1 demonstrates the overall architecture of Eucalyptus in data centre design. Eucalyptus CLIs can manage both AWS and its own private instances. Users can easily migrate instances from Eucalyptus to Amazon Elastic Cloud. Compute, storage and network is managed by the virtualisation layer. Instances are separated by hardware virtualisation. The following terminology is used by Eucalyptus.

Images: Any software module, configuration, application software or system software bundled and deployed in the Eucalyptus cloud is called a *Eucalyptus machine image (EMI)*.

Instances: When we run the image and use it, it becomes an instance. The controller will decide how much memory to allocate and provide all other resources.